



SAVING PRIVATE GRUB

Corporate hardware systems differ from their conventional desktop counterparts primarily because they have a number of interesting features. For example, they provide an ability to connect a virtual CD-ROM drive, and use a virtual monitor, just like a remote KVM. While this is not so important for desktop systems, it could be essential for large server farms. For example, let's imagine that you have accidentally damaged GRUB, and you're far away from your office. What should you do—ask someone to enter the server room and recover it, or can you try and fix it remotely? The answer is that you can easily recover GRUB from your current location!

Let's first understand the scenario, so that you know what I'm talking about. We have a couple of different projects hosted on a single branded (HP ProLiant 580) server whose configuration includes two RAID controllers, two physical Xeon processors and 24 GB of RAM — enough power for a dozen virtual machines. During the initial set-up of Debian, the server was configured to boot from and use only the first RAID controller, while the second controller was configured as RAID 5 and left intact.

After a while, some more projects were added on the

system, and it became clear that more space was needed; so it was decided to occupy the second RAID array as well. However, it was highly desirable to have it as RAID1+0 space; so a reconfiguration was done. However, after the first reboot, not a single ping was received from the server. What went wrong? Global loss of data, or something trickier?

My first response was to check whether GRUB was still good, or had been destroyed. If the latter, what would be my next step—a full Debian re-installation, together with all hosted virtual machines, and associated user data? From